

OmniSearch — MARL Training Experiments (HAPPO)

A record of the reinforcement-learning training runs for the wildfire search-and-rescue scenario: what was tried, episode budgets, results, and comparison against the non-learning baselines. Author: Oleksii Lavrenin (SeniorQAautomationEngineer). Generated 2026-06-13.

Goal & question

Central question: **can a trained MARL policy (HAPPO) match or beat hand-coded coordination heuristics** on the mission metric **survivor recall** (fraction of 5 survivors confirmed by a ground robot)? Drones scout (broad camera), ground robots confirm (precise, slow). Three hand-coded baselines provide the bar.

Evaluation conditions (all numbers below)

- **Terrain:** real cached terrain, ~1 km bbox (data/terrain_cache/malibu_creek_1km_128.npz), 128x128 grid.
- **Episode:** 1000 steps; recall = found/5 at episode end; mean over 5-10 seeds.
- **Reward-signal floors (opt-in, default 0.0):** drone_min_footprint = 0.15, ground_confirm_min = 0.20. On a real-km terrain the *physical* drone footprint (~0.033) and confirm range (~0.017) shrink so far that neither RL nor the baselines get a usable scout/confirm reward; the floors restore it. **Both HAPPO and the baselines are evaluated at the same floors** — comparisons are apples-to-apples. Defaults remain 0.0 (physical).

Non-learning baselines (the bar to beat)

Fixed hand-coded heuristics — no training. (Measured at floors 0.15 / 0.20, 1 km, 1000 steps.)

Baseline	How it works	scouted	recall
random	everyone moves randomly (control)	0.62	0.06
nearest_candidate	random drones; UGVs -> nearest scouted survivor	0.60	0.62
lawnmower	serpentine drone sweep; UGVs A* to nearest scouted	0.82	0.92

HAPPO training experiments (chronological)

Each run diagnosed and fixed a distinct bottleneck. Episodes x 500 steps = env steps.

#	Episodes	hrs	Key configuration	sct	rec	Finding / bottleneck
1	100	0.1	default 22 km terrain, no floors	0.2	0.07	Wrong (huge) terrain + no reward signal
2	1000	0.6	1 km, footprint floor 0.15	0.33	0.00	Cost-dominated reward -> do-nothing
3	2000	1.1	+entropy 0.08, search-reward	0.16	0.00	Reward-hacks shaping; action saturation
4	300*	0.2	+std_y 1.0, +entropy 0.25	0.47	0.00	Action saturation BROKEN; scouting up
5	3000	2.0	std1.0+ent0.25+search	0.56	0.00	Scouting up; ground never confirms
6	4000	2.6	+ground_confirm floor 0.12	0.36	0.16	Recall lifts off 0 first time
7	10000	8.6	same (non-recurrent)	0.33	0.10	Plateau - more budget does NOT help
8	6000	5.2	+coverage reward +recurrent (GRU)	0.62	0.12	Coverage breakthrough; scouting competitive
9	300*	0.2	+ground approach reward, confirm 0.20	0.56	0.20	Ground-confirm leg lifts recall to 0.20
10	6000	5.9	all fixes (coverage+recurrent+approach)	0.56	0.20	Plateaus at 0.20

* = short diagnostic run. **Total HAPPO training ~31 hours** across all runs (above + earlier 80k/400k experiments), plus behaviour-cloning/DAGger/evaluation. On CPU (Apple M-series), ~150-280 env-steps/sec; recurrent (GRU) runs are ~30% slower.

Learning from demonstration (cloning the lawnmower expert)

Pure RL plateaus at recall ~0.20. To approach the heuristic, clone its behaviour then RL-fine-tune.

Approach	scouted	recall	Note
Behaviour cloning (feedforward), no RL	0.74	0.30	Clones the sweep well; ground-confirm precision doesn't transfer
BC + RL fine-tune	0.46	0.36	Best learned policy; RL erodes the cloned sweep
Recurrent BC (GRU, BPTT)	0.74	0.38	BEST learned: memory captures multi-step navigation
Dagger (clone precision)	—	in progress	Re-label clone's own states with expert; fixes drift

Comparison: learned vs. non-learning

Policy	Trained?	recall
HAPPO — pure RL (best, all fixes)	yes	0.20
HAPPO — BC + RL fine-tune	yes	0.36
HAPPO — recurrent BC (best learned)	yes	0.38
random	no	0.06
nearest_candidate	no	0.62
lawnmower	no	0.92

Conclusion (honest)

- **HAPPO improved from a do-nothing collapse (0.00) to a functioning searcher (best learned recall 0.38, recurrent BC).** Drone scouting became competitive with the baselines (0.56-0.74 vs lawnmower 0.82).
- **It does NOT beat the hand-coded baselines** (nearest 0.62, lawnmower 0.92). Pure RL plateaus at ~0.20 — more episodes do not help (1k and 10k both ~0.10-0.20). Behaviour cloning lifts a learned policy to 0.30-0.38 (recurrent BC best).
- **Why:** six bottlenecks were diagnosed and fixed (footprint floor, action-saturation, coverage reward, recurrent memory, ground-confirm floor, ground-approach reward). The residual gap is the ground robots' A* approach-to-survivor precision, which neither RL nor feedforward BC reproduces. Recurrent BC + Dagger target exactly this and are running.
- **Scientific takeaway:** hand-coded coordination heuristics remain superior on mission recall; learned MARL matches the individual scouting sub-behaviour but not the full scout->confirm coordination at this budget.

Code & commits

Training scripts: `scripts/train_happo_smoke.py` (HAPPO via HARL; flags `--terrain-cache-path`, `--drone-min-footprint`, `--ground-confirm-min`, `--entropy-coef`, `--reward-search`, `--recurrent`, `--model-dir`), `train_bc_happo.py` (behaviour cloning), `train_dagger_happo.py` (Dagger). MAPPO/IPPO smoke scripts exist (BenchMARL) but their checkpoint->eval loaders are not wired, so they are not in the comparison.

- 34f27b6 — Add behaviour cloning from lawnmower + RL fine-tune for HAPPO
- dcd92e1 — Add ground directed-approach reward for the scout->confirm leg
- 384fa70 — Add coverage reward + recurrent policy: HAPPO learns systematic search
- 1c62f87 — Make HAPPO trainable: opt-in floors, anti-saturation, search reward
- 122acda — Floor drone scout footprint so survivors are detectable on real terrain

Note: all recall numbers are measured at the opt-in floors (0.15 / 0.20), not the physical defaults (0.0). Results are only comparable across the team at the same floor settings — a setting to standardize.